

VPPL SOLAR & WIND ENERGY SCHEDULING SYSTEM

INTELLIS WEEK-AHEAD LOGIC ARCHITECTURE

Metadata Field	Specification Details
Document ID	DOC-M04-S01-007
Document Title	Intellis_Week-Ahead_Logic_Architecture
Version	v1.0
Prepared by	Chaitanya Gore
Target Schedule Type	Week Ahead (WA, 7 Rolling Days / 672 Blocks)
Target Audience	Developers, Junior Developers / Interns
Classification	Official Production Technical Specification — Complete Implementation Blueprint
Last Updated	2026-10-08
Directory Name	schedule/, schedule/modules/week_ahead_forecast_solar/, schedule/modules/week_ahead_forecast_wind/
File Name	intellis_weekahead_solar_lambda.py, intellis_weekahead_wind_lambda.py, wa_solar_engine.py, wa_wind_engine.py

Document Revision History & Implementation Mandate

Revision	Release Date	Primary Author	Detailed Summary of Modifications & Architectural Scope
v1.0	2026-10-08	Chaitanya Gore	Initial Complete Production Specification for Week-Ahead Scheduling Architecture. Formulates end-to-end low-level algorithmic logic for both Solar and Wind Week-Ahead medium-range scheduling across a rolling 7-day horizon (Day T+1 through Day T+7, spanning 672 continuous 15-minute dispatch blocks). Key architectural pillars include: (1) Strategic elimination of generative LLM dependencies in favor of deterministic pure atmospheric physics and 168-hour synoptic Numerical Weather Prediction (NWP) multi-agency consensus; (2) Statutory compliance with Indian Electricity Grid Code (IEGC 2023) regional transmission coordination and corridor planning mandates; (3) Multi-day NREL SPA astronomical ephemeris and Ineichen clear-sky transposition across 672 continuous blocks; (4) 143-member NWP super-ensemble ingestion across ECMWF IFS, GFS, ICON, and GEM with continuous 168-hour physics-guided Kt clearness splines; (5) Multi-day Faiman and Sandia convective thermal module derating; (6) 21-slot diurnal ensembling with mandatory agency diversity capping (maximum 2 models per NWP agency) and Bayesian inverse-loss weighting; (7) DynamoDB multi-day planned maintenance outages and control window capacity scaling; (8) Multi-generator asset decomposition with dedicated column preservation across 672 blocks (e.g. ZTRIC, ENRICH, SHAHA); (9) 168-hour wind kinetic aerodynamic profiling, dynamic IEC 61400-12 air density normalization, Hellmann diurnal atmospheric boundary layer shear, and Jensen's-inequality-safe member-by-member power conversion; (10) Four physical guardrails; and (11) Canonical statutory 672-block CSV formatting and multi-path Amazon S3 dispatch.

Codebase Directory & Implementation File Tree Structure

The algorithmic logic, mathematical models, and data ingestion pipelines specified within this document are implemented across the following structured module repository hierarchy under schedule/. Each component is isolated for serverless AWS Lambda execution with zero cross-site dependency.

schedule/	
├─ intellis_weekahead_solar_lambda.py	[Centralized Solar WA Lambda Entrypoint]
├─ intellis_weekahead_wind_lambda.py	[Centralized Wind WA Lambda Entrypoint]
├─ deploy_weekahead_solar_lambda.py	[Solar WA Deployment & Rollback Script]
├─ deploy_weekahead_wind_lambda.py	[Wind WA Deployment & Rollback Script]
├─ modules/	
│ └─ week_ahead_forecast_solar/	
│ └─ wa_solar_engine.py	[Master Solar Week-Ahead Orchestrator]
│ └─ wa_sun_trajectory_engine.py	[672-Block Continuous Solar Ephemeris]

— wa_nwp_fetch_engine.py	[168h Synoptic Weather Fetch & Local Cache]
— wa_spline_kt_engine.py	[7-Day Multi-Spline Atmospheric Clearness]
— wa_power_derate_engine.py	[7-Day Maintenance Outage & Multi-Gen Engine]
— wa_formatter_s3.py	[Statutory 672-Block S3 CSV Formatter]
— week_ahead_forecast_wind/	
— wa_wind_engine.py	[Master Wind Week-Ahead Orchestrator]
— wa_wind_density_engine.py	[168h Synoptic Air Density Normalization]
— wa_wind_nwp_engine.py	[168h Hub Wind Splines & 28-Regime Ensembling]
— wa_wind_aerodynamic_engine.py	[Jensen's-Safe Power Mapping Across 672 Blocks]
— wa_wind_park_derate_engine.py	[7-Day Dynamic Park Derating & Availability]
— wa_wind_formatter_s3.py	[Statutory 672-Block S3 CSV Formatter]

Implementation File Mapping & Component Responsibilities:

Directory / Path	Code File Name	Architectural Role	Core Logic & Functions Implemented
schedule/	intellis_weekahead_solar_lambda.py	Solar WA Entrypoint	lambda_handler(), EventBridge Sunday/daily planning cron parsing, multi-plant 672-block batch loop.
schedule/	intellis_weekahead_wind_lambda.py	Wind WA Entrypoint	lambda_handler(), EventBridge planning cron parsing, 672 continuous 15-min block wind scheduling.
schedule/modules/week_ahead_forecast_solar/	wa_solar_engine.py	Master Solar WA	7-day rolling orchestration across ephemeris, 168h NWP, multi-splines, and outage scaling.
schedule/modules/week_ahead_forecast_solar/	wa_sun_trajectory_engine.py	Ephemeris Engine	Continuous 7-day solar geometry, solar zenith/azimuth, Ineichen clear-sky envelope.
schedule/modules/week_ahead_forecast_solar/	wa_nwp_fetch_engine.py	Synoptic NWP Fetch	168-hour multi-agency weather fetch (ECMWF, GFS, ICON, GEM), disk caching at /tmp/wa_weather_cache/.
schedule/modules/week_ahead_forecast_solar/	wa_spline_kt_engine.py	Clearness Splines	Daily atmospheric clearness index (Kt) calculation, 7 distinct daily cubic Hermite

			splines.
schedule/modules/week_ahead_forecast_solar/	wa_power_derate_engine.py	Outage & Multi-Gen	DynamoDB 7-day planned maintenance query, multi-generator column preservation across 672 blocks.
schedule/modules/week_ahead_forecast_solar/	wa_formatter_s3.py	S3 Lake Delivery	Statutory 672-row CSV assembly, S3 upload to intellis Weekhead solar/ for RLDC submission.
schedule/modules/week_ahead_forecast_wind/	wa_wind_engine.py	Master Wind WA	7-day rolling wind scheduling across synoptic weather, aerodynamic conversion, and park derating.
schedule/modules/week_ahead_forecast_wind/	wa_wind_density_engine.py	Air Density Engine	168-hour continuous thermodynamic air density calculation and barometric elevation correction.
schedule/modules/week_ahead_forecast_wind/	wa_wind_nwp_engine.py	Hub Wind Splines	Hellmann diurnal shear scaling to hub height, 28-regime diurnal slot ensembling across 7 days.
schedule/modules/week_ahead_forecast_wind/	wa_wind_aerodynamic_engine.py	Power Conversion	Jensen's-inequality-safe OEM power curve evaluation across 672 continuous blocks.
schedule/modules/week_ahead_forecast_wind/	wa_wind_park_derate_engine.py	Park Derate Engine	7-day dynamic park derating, scheduled turbine availability scaling, electrical loss modeling.
schedule/modules/week_ahead_forecast_wind/	wa_wind_formatter_s3.py	S3 Lake Delivery	Statutory 672-row Wind CSV assembly, S3 upload to intellis Weekhead wind/ for RLDC filing.
schedule/	deploy_weekahead_solar_lambda.py	Solar Deployment	Automated deployment, container image update, and rollback verification.

schedule/	deploy_weekahead_wind_lambda.py	Wind Deployment	Automated deployment, container image update, and rollback verification.
-----------	---------------------------------	-----------------	--

1. System Architecture & Week-Ahead Regulatory Governance Overview

The Intellis Week-Ahead Scheduling System is an enterprise-grade automated forecasting and grid planning platform engineered for utility-scale solar photovoltaic power stations and wind turbine generation fleets across India. The system complies strictly with the statutory mandates of the Indian Electricity Grid Code (IEGC 2023), Central Electricity Regulatory Commission (CERC) regulations, and State Electricity Regulatory Commission (SERC) frameworks, providing medium-range resource forecasts across a rolling 7-day forward operational horizon.

1.1 The Role of Week-Ahead Scheduling in Grid Planning & Market Operations

While Intraday scheduling addresses real-time deviations and Day-Ahead scheduling establishes the binding baseline for daily market clearing, Week-Ahead (WA) scheduling serves as the foundational medium-range projection tool for regional and national electrical grid operators. Regional Load Despatch Centres (RLDCs) and the National Load Despatch Centre (NLDC) utilize 7-day renewable generation forecasts for critical grid governance functions:

- **Transmission Corridor Congestion Management:** Evaluating inter-regional transmission corridor loading, anticipating potential transmission bottlenecks, and scheduling General Network Access (GNA) and Temporary General Network Access (T-GNA) power flows across national electrical highways.
- **Transmission Infrastructure Outage Planning:** Coordinating planned outages for ultra-high voltage (400 kV and 765 kV) substations, transmission lines, and grid transformers during anticipated low renewable generation windows to minimize curtailment risk.
- **Spinning Reserve & Resource Adequacy:** Assessing net demand requirements to schedule thermal coal and gas turbine spinning reserves and hydro pump storage dispatch to guarantee national grid frequency stability (49.90 Hz to 50.05 Hz).
- **Power Market & Bilateral Optimization:** Enabling commercial power producers, distribution companies (DISCOMs), and green traders to formulate bilateral power purchase agreements, participate in Green Day-Ahead (G-DAM) and Green Term-Ahead (G-TAM) forward markets, and optimize banking arrangements.

1.2 Rolling 7-Day / 672-Block Continuous Time Discretization

The Week-Ahead operational horizon spans exactly seven (7) consecutive calendar days, starting from Day T+1 through Day T+7. Each 24-hour day is discretized into ninety-six (96) standardized 15-minute time blocks. Consequently, the complete Week-Ahead schedule constitutes a continuous time series of exactly six hundred seventy-two (672) discrete dispatch blocks, indexed b in {1, 2, ..., 672}.

$$N_{WA} = 7 \text{ days} \times 96 \text{ blocks/day} = 672 \text{ blocks}, b \in \{1, 2, \dots, 672\}, t_{\text{start}}(b) = (b - 1) \times 15 \text{ min}$$

Where: Rolling 7-day Week-Ahead temporal discretization across 672 continuous dispatch blocks.

The mapping between global block index b and calendar day index d in $\{1, 2, \dots, 7\}$ and daily block index b_{day} in $\{1, 2, \dots, 96\}$ is evaluated deterministically as:

$$d(b) = \left\lfloor \frac{b-1}{96} \right\rfloor + 1, b_{\text{day}}(b) = ((b-1) \bmod 96) + 1, D(b) = D_{\text{today}} + d(b) \text{ days}$$

Where: Deterministic decomposition of 672-block index into calendar day and daily block index.

Horizon Day	Calendar Offset	Block Range (b)	Total Dispatch Blocks	Primary Synoptic Characteristics
Day 1	Day T+1	1 to 96	96 Blocks	High Synoptic Certainty; Microscale Cloud & Shear Boundary Coupling
Day 2	Day T+2	97 to 192	96 Blocks	High Frontal Tracking Certainty; Diurnal Thermal Gradient Transitions
Day 3	Day T+3	193 to 288	96 Blocks	Mesoscale Pressure Ridge / Trough Propagation; Air Density Shifts
Day 4	Day T+4	289 to 384	96 Blocks	Synoptic Weather Transition Window; Cloud Cover Spread Expansion
Day 5	Day T+5	385 to 480	96 Blocks	Ensemble Divergence Expansion; Multi-Agency Super-Ensemble Consensus
Day 6	Day T+6	481 to 576	96 Blocks	Medium-Range Jet Stream Positioning; Frontal Passage Probability
Day 7	Day T+7	577 to 672	96 Blocks	Medium-Range Climatological Anchoring; Macroscale Synoptic Regime Consensus

1.3 Strategic Elimination of Generative LLM in Medium-Range Scheduling

Similar to the Day-Ahead architecture, the Week-Ahead forecasting engine strictly eliminates all generative LLM invocations. Across a 168-hour forecast horizon, generative language models suffer from severe epistemic drift, lack multi-day convective conservation constraints, and cannot maintain physically consistent thermodynamic air density or solar ephemeris over 672 consecutive blocks. Instead, the Intellis Week-Ahead architecture relies exclusively on multi-agency Numerical Weather Prediction (NWP) super-ensembles, continuous physics-guided splines, and empirical diurnal transfer functions.

1.4 End-to-End Week-Ahead Pipeline Workflow & Orchestration

The Week-Ahead forecasting pipeline is automated via AWS EventBridge cron rules triggering daily or weekly runs:

- **Stage 1 — Automated Orchestration Gate:** AWS EventBridge triggers the Week-Ahead Lambda container via rule cron(30 5 * * ? *) corresponding to 11:00 AM IST.
- **Stage 2 — 7-Day Date Array Resolution:** The engine resolves execution date D_{today} and constructs the 7-day target date array $[D_{today} + 1 \dots D_{today} + 7]$ in IST.
- **Stage 3 — Multi-Day Maintenance Ingestion:** The system queries DynamoDB tables (vedanjay-control-windows) for multi-day scheduled maintenance outages, transformer overhauls, and transmission line shutdowns.
- **Stage 4 — Deterministic Physics Engines:** Astronomical solar ephemeris and clear-sky transposition (solar) or IEC 61400-12 dynamic air density (wind) are evaluated across all 672 blocks.
- **Stage 5 — 168-Hour NWP Super-Ensembling:** 168-hour NWP weather feeds are ingested across ECMWF IFS, GFS, ICON, and GEM, fitting continuous natural cubic splines to hourly data.
- **Stage 6 — Non-Linear Power Conversion:** Dynamic thermal cell convection (solar) or member-by-member non-linear power curve transformation (wind) generates candidate 672-block profiles.
- **Stage 7 — 21-Slot / 28-Regime Portfolio Consensus:** Medium-range historical MAE scores determine slot-wise champion models; agency diversity capping (max 2/agency) and Bayesian inverse-loss weights generate consensus.
- **Stage 8 — Multi-Generator Column Preservation:** Multi-generator sites (ZTRIC, ENRICH, SHAHA) allocate total generation across sub-assets proportionally, preserving dedicated columns across all 672 blocks.
- **Stage 9 — Physical Guardrails & Safety Clamping:** Physical guardrails enforce diurnal night zeroing, clear-sky caps, overcast bounds, and ramp rate continuity across the entire 7-day vector.
- **Stage 10 — Multi-Path S3 Enterprise Dispatch:** The finalized 672-block schedule is formatted into statutory CSV files and uploaded to Amazon S3 data lake storage.

2. Part A — Intellis Week-Ahead Solar Power Generation Scheduling System

Part A provides the exhaustive mathematical, physical, algorithmic, and software architecture specification for the Intellis Week-Ahead Solar Power Generation Scheduling System. Operating strictly on deterministic atmospheric physics, astronomical solar ephemeris, 168-hour multi-agency Numerical Weather Prediction (NWP) super-ensembles, dynamic PV cell thermal derating, multi-day planned maintenance outage scheduling, and multi-generator asset decomposition, this engine generates grid-compliant 672-block (7 rolling days @ 15-minute resolution) Week-Ahead generation schedules for utility-scale solar photovoltaic power stations across India without any generative LLM dependencies.

2.1 Global Variables, Astronomical Constants & 672-Block System Parameters

The following table defines the global system variables, physical constants, astronomical parameters, and operational coefficients governing the Week-Ahead solar scheduling engine across the entire 7-day (672-block) operational horizon:

Parameter Name	Symbol / Type	Standard Unit	Default / Valid Range	Functional Description & Context
HORIZON_DAYS	N_days (int)	Days	7 (Day T+1 to T+7)	Total number of continuous forward calendar days in the rolling Week-Ahead horizon.
DAILY_BLOCK_COUNT	N_daily_blocks (int)	Dimensionless	96	Number of standardized 15-minute dispatch intervals per 24-hour diurnal cycle.
TOTAL_WA_BLOCKS	N_WA (int)	Dimensionless	672	Total continuous 15-minute dispatch blocks in the complete 7-day schedule (7 x 96).
SCHEDULE_TRIGGER_IST	T_trigger (time)	Time (IST)	04:30:00 IST / 11:00:00 IST	Automated AWS EventBridge cron trigger time for Week-Ahead batch execution.
TIMEZONE_OPERATIONAL	TZ (str)	Timezone	Asia/Kolkata (UTC+05:30)	Statutory reference timezone for Indian electricity grid operations.
SOLAR_CONSTANT	I_0 (float)	W / m ²	1361.2	Extraterrestrial solar irradiance at mean Earth-Sun orbital distance (NREL standard).
STC_REFERENCE_IRRADIANCE	G_STC (float)	W / m ²	1000.0	Standard Test Conditions (STC) reference irradiance for PV module nameplate rating.
STC_CELL_TEMPERATURE	T_STC (float)	Celsius	25.0	Standard Test Conditions reference photovoltaic semiconductor junction temperature.
TEMPERATURE_COEFF_F_PMP	gamma_pmp (float)	1 / Celsius	-0.0035 to -0.0042	Photovoltaic module maximum power temperature derating coefficient.
NOCT_TEMPERATURE	T_NOCT (float)	Celsius	45.0 to 48.0	Nominal Operating Cell Temperature under 800 W/m ² , 20 deg C ambient, 1 m/s wind.
FAIMAN_CONVECTION_U0	u_0 (float)	W / (m ² K)	25.0 to 29.0	Faiman thermal convective heat transfer constant for open-rack ground mount.
FAIMAN_CONVECTION_U1	u_1 (float)	W / (m ³ K s)	1.2 to 1.5	Faiman thermal wind convective cooling velocity coefficient.
ELEVATION_CUTOFF_DEGREES	alpha_min (float)	Degrees	3.0	Solar elevation angle below which solar irradiance is forcibly clamped to 0.0.
ZENITH_NIGHT_THRESHOLD	theta_z_night	Degrees	89.5	Solar zenith angle threshold above

SHOLD	(float)			which night zeroing is unconditionally enforced.
SYNOPTIC_HOURS_HORIZON	H_synoptic (int)	Hours	168	Total continuous synoptic weather hours ingested from NWP APIs (7 days x 24 hours).
NWP_ENSEMBLE_MEMBERS	M (int)	Dimensionless	143	Total ensemble weather prediction members ingested across ECMWF, GFS, ICON, GEM.
DIURNAL_SLOTS_PER_DAY	N_slots_day (int)	Dimensionless	3	Operational diurnal slots per day: Morning (06-10), Peak (10-15), Afternoon (15-19).
TOTAL_WA_SLOTS	N_slots_total (int)	Dimensionless	21	Total operational diurnal evaluation slots across the 7-day horizon (7 x 3).
MAX_MODELS_PER_AGENCY	K_agency (int)	Dimensionless	2	Agency diversity cap: maximum candidate models selected from any single NWP agency.
SLOT_CHAMPION_COUNT	K_slot (int)	Dimensionless	6	Number of diverse champion weather models selected per diurnal operational time slot.
INVERTER_EFFICIENCY	eta_inv (float)	Fraction	0.985 (98.5%)	Standard utility-scale central inverter DC-to-AC conversion efficiency.
MAX_RAMP_FRACTION	R_max (float)	Fraction	0.10 (10% P _{ac})	Maximum allowable active power ramp rate change between adjacent 15-minute blocks.

2.2 End-to-End Solar Week-Ahead Execution Flow

Upon scheduled invocation by AWS EventBridge, the Intellis Week-Ahead Solar Scheduling System executes the following nine (9) deterministic computational modules in strict sequential dependency:

- **Module S1: 7-Day Rolling Horizon Temporal Resolution & Asset Ingestion:** Resolve execution date D_{today} and construct the 7-day target date vector $[D_{today} + 1 \dots D_{today} + 7]$ in Asia/Kolkata timezone; load plant hardware parameters (AC capacity, DC capacity, inverter stations, tilt, azimuth) for all registered solar plants.
- **Module S2: 672-Block Multi-Day NREL SPA Ephemeris & Ineichen Clear-Sky Transposition:** Compute high-precision astronomical ephemeris (solar zenith, azimuth, optical air mass) for all 672 block midpoints (+7.5 minutes); evaluate Ineichen clear-sky radiation triplet (GHI, DNI, DHl); transpose to plane-of-array (POA) via Hay-Davies/Perez transposition; establish deterministic clear-sky ceilings across all 7 days.
- **Module S3: 168-Hour Multi-Agency NWP Super-Ensemble Ingestion & Multi-Day Kt Clearness Splines:** Ingest 168 hours of synoptic weather forecasts across 143 ensemble members from ECMWF IFS,

GFS, ICON, and GEM; compute hourly atmospheric clearness index $K_t(H)$; fit daily natural cubic splines for each of the 7 days; reconstruct continuous 672-block Global Tilted Irradiance (GTI) matrix.

- **Module S4: Dynamic Thermal Cell Convection & Multi-Day Base Power Simulation:** Interpolate 168-hour ambient temperatures to 672 blocks; simulate dynamic PV cell junction temperature $T_{cell}(b)$ via Sandia/Faiman convection; compute thermal derate factor $\eta_{thermal}(b)$; generate 143 candidate AC generation profiles clamped to AC interconnect capacity.
- **Module S5: 21-Slot Diurnal Ensembling, Agency Diversity Capping & Consensus Synthesis:** Partition the 7-day horizon into 21 operational time slots (3 slots/day x 7 days); score ensemble members against medium-range loss functions; enforce mandatory agency diversity capping (maximum 2 models per NWP agency); compute Bayesian inverse-loss weights; apply cosine cross-fading across slot boundaries; synthesize consensus power curve.
- **Module S6: DynamoDB 7-Day Planned Maintenance Outages & Control Window Scaling:** Query DynamoDB table (vedanjay-control-windows) for multi-day scheduled maintenance outages across the 7-day window; map outage time windows to 672 block indices; apply full shutdown and partial DC string capacity derates.
- **Module S7: Multi-Generator Solar Asset Decomposition & 672-Block Column Allocation:** For multi-generator solar parks (e.g. ZTRIC, ENRICH, SHAHA), allocate total park schedule across individual generator entities proportionally to their rated capacity; preserve dedicated asset columns across all 672 blocks in memory and output CSVs.
- **Module S8: Four Physical Guardrails & 672-Block Safety Clamping:** Screen the raw 672-block schedule against 4 strict physical guardrails: (1) Astronomical night zeroing, (2) Clear-sky envelope ceiling (105% P_{cs_ac}), (3) Overcast optical cloud floor, and (4) Inverter active power ramp rate continuity filter.
- **Module S9: Canonical Statutory 672-Block CSV Formatting & Amazon S3 Dispatch:** Format the finalized schedule into statutory CSV files with 672 rows, continuous or resetting block index, 'From', 'To', 'SCH_MW', and 'AvC_MW'; upload to Amazon S3 data lake paths for RLDC/SLDC grid submission and operational dashboards.

2.3 Module S1: 7-Day Rolling Horizon Temporal Resolution & Asset Configuration

Module Context: Week-Ahead scheduling operates on a medium-range rolling 7-day planning window. The engine deterministically resolves execution timing in Indian Standard Time (IST, UTC+05:30) and constructs the 7-day target date array:

$$D_{today} = \text{date}(\text{now}_{IST}), D_{target}(d) = D_{today} + d \text{ for } d \in \{1, 2, \dots, 7\}$$

Where: Dynamic 7-day calendar date array resolution in Asia/Kolkata timezone.

Continuous 672-Block Time Vector Construction: Each day d in $\{1, \dots, 7\}$ contains exactly 96 15-minute dispatch intervals. The global block index b in $\{1, 2, \dots, 672\}$ maps to calendar day d and daily block index b_{day} as:

$$d(b) = \left\lfloor \frac{b-1}{96} \right\rfloor + 1, b_{day}(b) = ((b-1) \bmod 96) + 1, t_{mid}(b) = t_{start}(b) + 7.5 \text{ min}$$

Where: Mapping of 672 global blocks to calendar day, daily block index, and block midpoint timestamp.

Plant Hardware Profile Ingestion: For each registered solar site, the engine loads hardware specifications including AC interconnect capacity P_CAP_AC , installed DC nameplate capacity P_CAP_DC , array tilt angle β , array azimuth angle γ_m (relative to true south), inverter efficiency η_{inv} , BOS loss factor η_{bos} , and sub-generator asset topologies.

2.4 Module S2: 672-Block Multi-Day NREL SPA Ephemeris & Ineichen Clear-Sky Transposition

Module Context: The physical upper envelope of solar power generation across the 7-day horizon is governed by celestial mechanics. The engine computes astronomical ephemeris at each of the 672 block midpoints $t_mid(b)$ using the NREL Solar Position Algorithm (SPA).

Multi-Day Solar Declination & Equation of Time: For each calendar day d in $\{1, \dots, 7\}$, day of year $n(d)$ is evaluated to compute solar declination $\delta(d)$ and Equation of Time $EoT(d)$:

$$n(d) = \text{day_of_year}(D_{\text{target}}(d)), \delta(d) = 23.45^\circ \sin\left(\frac{360^\circ}{365}(284 + n(d))\right)$$

Where: Solar declination angle δ evaluated across each of the 7 calendar days.

$$B = \frac{360^\circ}{365}(n(d) - 81), EoT(d) = 9.87 \sin(2B) - 7.53 \cos(B) - 1.5 \sin(B) [\text{minutes}]$$

Where: Equation of Time (EoT) in minutes accounting for orbital eccentricity and axial obliquity.

Local Solar Time & Zenith Angle: At each block midpoint b in $\{1, \dots, 672\}$, local solar time $t_solar(b)$ and hour angle $\omega(b)$ are evaluated:

$$t_{\text{solar}}(b) = t_{\text{local}}(b) + \frac{4(\lambda - \lambda_{\text{ref}}) + EoT(d(b))}{60}, \omega(b) = (t_{\text{solar}}(b) - 12.0) \times 15^\circ/\text{hr}$$

Where: Local solar time and hour angle ω where λ is site longitude and $\lambda_{\text{ref}} = 82.5 \text{ deg E}$ for IST.

$$\cos(\theta_z(b)) = \sin(\phi) \sin(\delta(d(b))) + \cos(\phi) \cos(\delta(d(b))) \cos(\omega(b))$$

Where: Solar zenith angle $\theta_z(b)$ evaluated across all 672 block midpoints where ϕ is site latitude.

Ineichen Clear-Sky Radiance & Hay-Davies/Perez Transposition: The Ineichen clear-sky model evaluates Global Horizontal Irradiance $GHI_cs(b)$, Direct Normal Irradiance $DNI_cs(b)$, and Diffuse Horizontal Irradiance $DHI_cs(b)$ using Linke turbidity TL . Plane-of-Array Irradiance $POA_cs(b)$ is transposed onto the tilted PV collector plane:

$$POA_{cs}(b) = DNI_{cs}(b) \cos(\theta_i(b)) + DHI_{cs}(b) \left[\frac{DNI_{cs}(b) \cos(\theta_i(b))}{I_0 \cos(\theta_z(b))} + \left(1 - \frac{DNI_{cs}(b)}{I_0} \right) \left(\frac{1 + \cos(\beta)}{2} \right) \right] +$$

Where: Hay-Davies clear-sky Plane-of-Array (POA) transposition across all 672 blocks where θ_i is incidence angle, β is tilt, ρ_g is albedo.

Astronomical Night Zeroing: For all blocks where solar zenith angle $\theta_z(b) \geq 89.5$ degrees or solar elevation $\alpha(b) \leq 3.0$ degrees, $POA_{cs}(b)$ and $GHI_{cs}(b)$ are unconditionally clamped to exactly 0.0 W/m^2 .

2.5 Module S3: 168-Hour Multi-Agency NWP Super-Ensemble Ingestion & Multi-Day Kt Clearness Splines

Module Context: Numerical Weather Prediction across a 7-day medium-range horizon is inherently uncertain. To achieve maximum forecast resilience, the engine ingests 168 continuous hours of weather data across 143 ensemble members from four leading meteorological agencies: ECMWF (European Centre for Medium-Range Weather Forecasts IFS), NCEP (National Centers for Environmental Prediction GFS), DWD (Deutscher Wetterdienst ICON), and ECCC (Environment and Climate Change Canada GEM).

Atmospheric Clearness Index Calculation: Direct linear interpolation of NWP solar irradiance creates severe step-change artifacts and violates astronomical sunrise/sunset constraints. To guarantee physical continuity, the engine transforms raw NWP Global Horizontal Irradiance into the dimensionless atmospheric clearness index K_t at each synoptic hour H in $\{0, 1, \dots, 167\}$:

$$K_t(H) = \frac{GHI_{nwp}(H)}{\max(10.0, GHI_{cs_hourly}(H))}, H \in \{0, 1, \dots, 167\}$$

Where: Atmospheric clearness index K_t computed at hourly synoptic timesteps across the 168-hour forecast horizon.

Daily Natural Cubic Spline Interpolation: To eliminate inter-day leakage while ensuring smooth intra-day variation, the engine fits a natural cubic spline $S_d(t)$ independently for each calendar day d in $\{1, \dots, 7\}$ over t in $[0, 23]$ hours:

$$S'_d(0) = 0, S'_d(23) = 0, S_d(t) = a_i + b_i(t - t_i) + c_i(t - t_i)^2 + d_i(t - t_i)^3$$

Where: Natural cubic spline formulation fitted across 24 hourly K_t nodes for each calendar day d .

Reconstruction of 672-Block POA Matrix: The fitted spline $S_d(t)$ is evaluated at the 96 block midpoints for each day, clipped to the physical range $[0.0, 1.25]$, and multiplied by the theoretical clear-sky POA profile to generate the multi-member POA matrix of shape $(M, 672)$:

$$POA_m(b) = \text{clip}(S_{d(b),m}(t_{rel}(b)), 0.0, 1.25) \times POA_{cs}(b), m \in \{1, \dots, M\}, b \in \{1, \dots, 672\}$$

Where: Reconstruction of continuous 672-block Plane-of-Array irradiance matrix across M ensemble members.

2.6 Module S4: Dynamic Thermal Cell Convection & Multi-Day Base Power Simulation

Module Context: Photovoltaic conversion efficiency degrades significantly with increasing semiconductor junction temperature. In Indian climatic zones (e.g. Gujarat, Rajasthan, Telangana, Madhya Pradesh), cell temperatures frequently exceed 60 deg C during peak hours.

Ambient Temperature Interpolation: Hourly 2-meter ambient temperatures $T_{\text{amb}}(H)$ from NWP feeds across 168 hours are interpolated to all 672 15-minute dispatch blocks via linear interpolation:

$$T_{\text{amb}}(b) = T_{\text{amb}}(H(b)) + \frac{b_{\text{hour}}(b) - 1}{4} [T_{\text{amb}}(H(b) + 1) - T_{\text{amb}}(H(b))]$$

Where: Continuous 15-minute ambient temperature vector across 672 blocks.

PV Cell Junction Temperature Simulation: Using the Sandia / Faiman convective heat transfer formulation, the cell temperature $T_{\text{cell}}(b)$ is computed as:

$$T_{\text{cell},m}(b) = T_{\text{amb}}(b) + \frac{\text{POA}_m(b)}{800.0} \times (T_{\text{NOCT}} - 20.0)$$

Where: Photovoltaic cell junction temperature simulation across all 672 blocks.

Thermal Derating Factor & Active Power Generation: Maximum power temperature derating factor $\eta_{\text{thermal}}(b)$ is evaluated and applied to DC nameplate capacity $P_{\text{CAP_DC}}$. Inverter conversion efficiency η_{inv} (98.5%) and BOS losses η_{bos} are applied, and the output is clamped to AC interconnect capacity $P_{\text{CAP_AC}}$:

$$\eta_{\text{thermal},m}(b) = \text{clip}(1.0 + \gamma_{\text{pmp}}(T_{\text{cell},m}(b) - 25.0), 0.70, 1.10)$$

Where: Semiconductor maximum power thermal derate factor bounded between 70% and 110%.

$$P_{\text{dc},m}(b) = P_{\text{CAP_DC}} \times \left(\frac{\text{POA}_m(b)}{1000.0} \right) \times \eta_{\text{thermal},m}(b) \times \eta_{\text{bos}}$$

Where: Gross DC active power generation across 672 blocks for ensemble member m .

$$P_{\text{ac},m}(b) = \min(P_{\text{dc},m}(b) \times \eta_{\text{inv}}, P_{\text{CAP_AC}})$$

Where: Net AC active power generation clipped to plant AC interconnect rating.

2.7 Module S5: 21-Slot Diurnal Ensembling, Agency Diversity Capping & Consensus Synthesis

Module Context: Single deterministic weather forecasts diverge substantially beyond Day T+3. To maximize forecast skill and mitigate multi-day synoptic drift, the engine employs a structured 21-slot diurnal ensembling framework across the 7-day horizon.

21-Slot Diurnal Horizon Partitioning: Each of the 7 calendar days is partitioned into three (3) distinct operational time slots:

- **Morning Ramp Slot (S_morn):** Blocks 25 to 40 (06:00 to 10:00 IST) — Characterized by rapid solar elevation ramp-up and boundary layer dissipation.
- **Peak Plateau Slot (S_peak):** Blocks 41 to 60 (10:00 to 15:00 IST) — Maximum solar irradiance, peak thermal cell derating, and convective cloud formation.
- **Afternoon Ramp Slot (S_aftn):** Blocks 61 to 74 (15:00 to 18:30 IST) — Afternoon thermal dissipation, declining solar elevation, and evening ramp-down.

Global Slot Index	Horizon Day	Diurnal Phase	Local Time (IST)	Daily Block Range	Global Block Range
Slot 1	Day T+1	Morning Ramp	06:00 to 10:00 IST	Blocks 25 to 40	Blocks 25 to 40
Slot 2	Day T+1	Peak Plateau	10:00 to 15:00 IST	Blocks 41 to 60	Blocks 41 to 60
Slot 3	Day T+1	Afternoon Ramp	15:00 to 18:30 IST	Blocks 61 to 74	Blocks 61 to 74
Slot 4	Day T+2	Morning Ramp	06:00 to 10:00 IST	Blocks 25 to 40	Blocks 121 to 136
Slot 5	Day T+2	Peak Plateau	10:00 to 15:00 IST	Blocks 41 to 60	Blocks 137 to 156
Slot 6	Day T+2	Afternoon Ramp	15:00 to 18:30 IST	Blocks 61 to 74	Blocks 157 to 170
Slot 7 to 9	Day T+3	Morning / Peak / Aftn	06:00 to 18:30 IST	Blocks 25 to 74	Blocks 217 to 266
Slot 10 to 12	Day T+4	Morning / Peak / Aftn	06:00 to 18:30 IST	Blocks 25 to 74	Blocks 313 to 362
Slot 13 to 15	Day T+5	Morning / Peak / Aftn	06:00 to 18:30 IST	Blocks 25 to 74	Blocks 409 to 458
Slot 16 to 18	Day T+6	Morning / Peak / Aftn	06:00 to 18:30 IST	Blocks 25 to 74	Blocks 505 to 554
Slot 19 to 21	Day T+7	Morning / Peak / Aftn	06:00 to 18:30 IST	Blocks 25 to 74	Blocks 601 to 650

Agency Diversity Capping ($K_{\text{agency}} \leq 2$): For each slot, candidate weather models are ranked based on historical validation skill. To prevent single-agency dominance (e.g. ECMWF over-fitting), the selection algorithm enforces a strict quota: a maximum of two (2) models may be chosen from any single meteorological agency. If three ECMWF models rank in the top 6, the third model is disqualified in favor of the next highest-ranking model from GFS, ICON, or GEM.

$$N_{\text{selected}}(\text{Agency}_a, s) \leq K_{\text{agency}} = 2 \text{ for all } a \in \{\text{ECMWF, GFS, ICON, GEM}\}, \text{ for all } s \in \{1, \dots, 21\}$$

Where: Mathematical agency diversity capping constraint across all 21 operational diurnal slots.

Bayesian Inverse-Loss Weighting & Consensus Synthesis: Within each slot s , selected champion models k in $\{1, \dots, K_{\text{slot}}\}$ receive normalized weights $w_k(s)$ inversely proportional to their squared historical validation error:

$$w_k(s) = \frac{\text{MAE}_k(s)^{-2}}{\sum_{j=1}^{K_{\text{slot}}} \text{MAE}_j(s)^{-2}}, P_{\text{slot}}(b) = \sum_{k=1}^{K_{\text{slot}}} w_k(s) \times P_{\text{ac},k}(b)$$

Where: Bayesian inverse-loss weighted consensus formulation within operational slot s .

Robust Median Fallback: In operational environments where historical ground truth validation data is temporarily unavailable or sparse for medium-range horizons (e.g. Days T+4 to T+7), the engine activates a robust non-parametric median consensus fallback:

$$P_{\text{consensus}}(b) = \text{median}(\{P_{\text{ac},m}(b)\}_{m=1}^M) \text{ for all } b \in \{1, 2, \dots, 672\}$$

Where: Robust non-parametric median consensus fallback across all M ensemble members for block b .

2.8 Module S6: DynamoDB 7-Day Planned Maintenance Outages & Control Window Scaling

Module Context: Utility-scale solar plants undergo planned inverter overhauls, transformer preventative maintenance, and switchyard shutdowns during medium-range horizons. The engine integrates with AWS DynamoDB (table vedanjay-control-windows) to ingest and apply planned maintenance outages across the complete 7-day operational window $[D_{\text{today}} + 1 .. D_{\text{today}} + 7]$.

Maintenance Window Parsing & Block Mapping: The engine queries DynamoDB for items matching the site ID and target dates. Each maintenance record specifies start timestamp T_{start} , end timestamp T_{end} , outage type (FULL_SHUTDOWN or PARTIAL_DERATE), and affected capacity fraction delta_C . Start and end times are mapped to continuous 672-block indices:

$$b_{\text{start}} = \left\lfloor \frac{T_{\text{start}} - T_0}{15 \text{ min}} \right\rfloor + 1, b_{\text{end}} = \left\lceil \frac{T_{\text{end}} - T_0}{15 \text{ min}} \right\rceil$$

Where: Mapping of maintenance window timestamps to continuous 672-block index range.

Capacity Derating Formulation: The active available capacity $P_{\text{avail_cap}}(b)$ and schedule derate are evaluated deterministically:

$$P_{\text{avail_cap}}(b) = P_{\text{CAP_AC}} \times \left(1.0 - \sum_{k \in \mathcal{M}(b)} \Delta C_k \right), P_{\text{maint}}(b) = \min(P_{\text{consensus}}(b), P_{\text{avail_cap}}(b))$$

Where: Maintenance-derated active generation capacity where $\mathcal{M}(b)$ is the set of active maintenance outages during block b .

2.9 Module S7: Multi-Generator Solar Asset Decomposition & 672-Block Column Allocation

Module Context: Several utility-scale solar parks operate as pooled injection nodes comprising multiple distinct generator legal entities or power purchase agreements (PPAs). For example, the SHAHA site comprises SIDDEHESH (15 MW), PRANAV (15 MW), and LOKGREENB2 (10 MW); ZTRIC comprises ZTRIC_1 and ZTRIC_2; and ENRICH comprises multiple distinct feeder units.

Proportional Asset Capacity Allocation: The engine allocates the total park schedule $P_{\text{total}}(b)$ across individual generator entities j in $\{1, \dots, N_{\text{assets}}\}$ based on their validated nameplate interconnect capacity C_j :

$$f_j = \frac{C_j}{\sum_{i=1}^{N_{\text{assets}}} C_i}, P_{\text{asset},j}(b) = P_{\text{total}}(b) \times f_j \text{ for all } b \in \{1, 2, \dots, 672\}$$

Where: Proportional active power allocation across multi-generator assets for block b .

Dedicated Column Preservation: In contrast to basic systems that flatten multi-generator outputs into a single aggregated column, the Intellis Week-Ahead engine preserves dedicated time-series columns for each sub-asset across all 672 dispatch blocks. The output DataFrame schema enforces: [Block_No, Date, Time_Interval, TOTAL_MW, ASSET_1_MW, ASSET_2_MW, ..., ASSET_N_MW].

2.10 Module S8: Four Physical Guardrails & 672-Block Safety Clamping

Module Context: To guarantee that no physically impossible values are dispatched to grid operators, the 672-block generation vector undergoes four (4) sequential physical guardrails:

- **Guardrail 1 — Astronomical Night Zeroing:** For all blocks where solar zenith angle $\theta_z(b) \geq 89.5$ degrees or theoretical clear-sky POA_{cs}(b) $\leq 1.0 \text{ W/m}^2$, active power is unconditionally clamped to exactly 0.00 MW: $P_{\text{guard}}(b) = 0.0$.
- **Guardrail 2 — Clear-Sky Ceiling Envelope Clamp:** Photovoltaic power cannot physically exceed the maximum possible clear-sky radiation envelope. Active generation is clamped to 105% of the clear-sky AC conversion curve $P_{\text{cs_ac}}(b)$ to accommodate transient cloud-edge irradiance enhancement: $P_{\text{guard}}(b) \leq 1.05 * P_{\text{cs_ac}}(b)$.
- **Guardrail 3 — Overcast Optical Cloud Floor:** Under 100% overcast cloud cover, diffuse irradiance maintains a physical minimum floor of 8% to 12% of clear-sky irradiance: $P_{\text{guard}}(b) \geq 0.08 * P_{\text{cs_ac}}(b)$ for daytime blocks.

- **Guardrail 4 — Inverter Active Power Ramp Rate Filter:** Inverter physical ramping capabilities limit maximum power rate-of-change between adjacent 15-minute blocks to 10% of plant AC capacity: $|P_{\text{guard}}(b) - P_{\text{guard}}(b - 1)| \leq 0.10 * P_{\text{CAP_AC}}$.

$$P_{\text{final}}(b) = \min(\max(P_{\text{ramp}}(b), 0.0), 1.05 \times P_{\text{cs_ac}}(b), P_{\text{avail_cap}}(b))$$

Where: Master physical guardrail synthesis across all 672 continuous dispatch blocks.

2.11 Module S9: Canonical Statutory 672-Block CSV Formatting & Amazon S3 Dispatch

Module Context: The finalized 672-block Week-Ahead schedule is formatted into statutory CSV structures compliant with RLDC / SLDC transmission scheduling guidelines. The engine supports two statutory block enumeration formats:

- **Format A — Daily Resetting Block Index (1 to 96 per Day):** S. No resets from 1 to 96 at the start of each calendar day (Format A, preferred by state load despatch centres).
- **Format B — Continuous 672-Block Index (1 to 672):** S. No runs continuously from 1 to 672 across the entire 7-day horizon (Format B, preferred by regional transmission planners).

S. No	From	To	SCH_MW	AvC_MW
1	2026-10-09 00:00	2026-10-09 00:15	0.00	50.00
28	2026-10-09 06:45	2026-10-09 07:00	4.12	50.00
48	2026-10-09 11:45	2026-10-09 12:00	43.85	50.00
72	2026-10-09 17:45	2026-10-09 18:00	2.40	50.00
96	2026-10-09 23:45	2026-10-10 00:00	0.00	50.00
1 (97)	2026-10-10 00:00	2026-10-10 00:15	0.00	50.00
...	...	...	...	...
96 (672)	2026-10-15 23:45	2026-10-16 00:00	0.00	50.00

Amazon S3 Multi-Path Enterprise Dispatch: Upon CSV serialization, the engine uploads the 672-row schedule to Amazon S3 data lake storage across standardized hierarchical paths:

- **Test Data Lake Path:** s3://vedanjay-schedules-test-608744602858/intellis Weekhead solar/{SITE}/{TODAY_DATE}/{PLANT}_{START_DATE}_weekahead_WA.csv
- **Production Data Lake Path:** s3://vedanjay-schedules-prod/intellis Weekhead solar/{SITE}/{TODAY_DATE}/{PLANT}_{START_DATE}_weekahead_WA.csv
- **Raw NWP Ensemble Archive:** s3://vedanjay-schedules-prod/raw_archives/week Ahead_solar/{YEAR}/{MONTH}/{SITE}_{START_DATE}_WA_672_raw.json

3. Part B — Intellis Week-Ahead Wind Power Generation Scheduling System

Part B provides the exhaustive aerodynamic, thermodynamic, mathematical, and algorithmic specification for the Intellis Week-Ahead Wind Power Generation Scheduling System. Operating strictly on fluid dynamics, atmospheric boundary layer shear, 168-hour multi-agency Numerical Weather Prediction (NWP) hub-height super-ensembles, dynamic IEC 61400-12 air density normalization, Jensen's-inequality-safe member-by-member non-linear power curve transformation, and empirical park derating, this engine generates grid-compliant 672-block (7 rolling days @ 15-minute resolution) Week-Ahead generation schedules for utility-scale wind turbine farms across India without stochastic LLM hallucinations.

3.1 Global Aerodynamic Variables, Turbine Constants & 672-Block System Parameters

The following table defines the global aerodynamic variables, physical constants, thermodynamic properties, and turbine operational parameters governing the Week-Ahead wind scheduling engine across the entire 7-day (672-block) horizon:

Parameter Name	Symbol / Type	Standard Unit	Default / Valid Range	Functional Description & Context
HORIZON_DAYS	N_days (int)	Days	7 (Day T+1 to T+7)	Total number of continuous forward calendar days in the rolling Week-Ahead horizon.
DAILY_BLOCK_COUNT	N_daily_blocks (int)	Dimensionless	96	Number of standardized 15-minute dispatch intervals per 24-hour diurnal cycle.
TOTAL_WA_BLOCKS	N_WA (int)	Dimensionless	672	Total continuous 15-minute dispatch blocks in the complete 7-day schedule (7 x 96).
SCHEDULE_TRIGGER_IST	T_trigger (time)	Time (IST)	04:30:00 IST / 11:00:00 IST	Automated AWS EventBridge cron trigger time for Week-Ahead batch execution.
TIMEZONE_OPERATIONAL	TZ (str)	Timezone	Asia/Kolkata (UTC+05:30)	Statutory reference timezone for Indian electricity grid operations.
STANDARD_AIR_DENSITY	rho_0 (float)	kg / m ³	1.225	Standard sea-level dry air density at 1013.25 hPa and 15.0 deg C (IEC 61400-12).
SPECIFIC_GAS_CONSTANT	R_spec (float)	J / (kg K)	287.058	Specific gas constant for dry atmospheric air in ideal gas thermodynamic calculations.
REFERENCE_HUB_HEIGHT	z_hub (float)	Meters (m)	120.0 to 140.0	Physical centerline turbine hub height above ground level (AGL).
CUT_IN_WIND_SPEED	v_cut_in (float)	m / s	3.0	Minimum effective hub-height

				wind speed required for aerodynamic rotor torque generation.
RATED_WIND_SPEED	v Rated (float)	m / s	10.0 to 11.5	Wind speed at which turbine reaches maximum electrical nameplate rated capacity.
CUT_OUT_WIND_SPEED	v_cut_out (float)	m / s	20.0 to 25.0	Wind speed above which emergency pitch-to-feather shutdown is triggered for structural safety.
RE_CUT_IN_WIND_SPEED	v_re_cut_in (float)	m / s	18.0 to 22.0	Hysteresis re-cut-in wind speed threshold following a high-wind cut-out event.
HELLMANN_EXPONENT_DAY	alpha_day (float)	Dimensionless	0.10 to 0.14	Daytime unstable/convective boundary layer wind shear power-law exponent.
HELLMANN_EXPONENT_NIGHT	alpha_night (float)	Dimensionless	0.22 to 0.28	Nocturnal stable boundary layer wind shear power-law exponent.
SYNOPTIC_HORIZON_HOURS	H_synoptic (int)	Hours	168	Total continuous synoptic weather hours ingested from NWP APIs (7 days x 24 hours).
NWP_ENSEMBLE_MEMBERS	M (int)	Dimensionless	143	Total ensemble weather prediction members ingested across ECMWF, GFS, ICON, GEM.
REGIMES_PER_DAY	N_regimes_day (int)	Dimensionless	4	Operational wind regimes per day: Nocturnal Jet, Morning Dissipation, Thermal Convection, Evening Shear.
TOTAL_WA_REGIMES	N_regimes_total (int)	Dimensionless	28	Total operational diurnal wind regimes across the 7-day horizon (7 x 4).
WAKE_DERATE_FACTOR	eta_wake (float)	Fraction	0.950 (95.0%)	Empirical Jensen wake interaction loss derate factor across turbine arrays.
ELECTRICAL_LOSS_FACTOR	eta_elec (float)	Fraction	0.970 (97.0%)	Internal 33 kV collector cable impedance, transformer iron/copper losses, and substation losses.
AVAILABILITY_FACTOR	eta_avail (float)	Fraction	0.966 (96.6%)	Fleet operational availability factor accounting for minor curtailment and blade contamination.

COMBINED_PARK_DERATE	eta_park (float)	Fraction	0.890 (89.0%)	Master combined wind park derate factor: eta_park = eta_wake * eta_elec * eta_avail.
CONTINUOUS_OPERATION	Is24h (bool)	Boolean	True	Wind turbines operate 24 hours continuously; astronomical night zeroing is STRICTLY PROHIBITED.

3.2 End-to-End Wind Week-Ahead Execution Flow

Upon scheduled invocation by AWS EventBridge, the Intellis Week-Ahead Wind Scheduling System executes the following nine (9) deterministic computational modules in strict sequential dependency:

- **Module W1: 7-Day Horizon Discretization & Wind Hardware Profile Ingestion:** Resolve execution date D_{today} and construct the 7-day target date array $[D_{today} + 1 .. D_{today} + 7]$ in IST; load turbine hardware profiles (JEWLI Siemens Gamesa SG 3.6-145 100.8 MW, JGBPL Envision EN182-5.0 50.0 MW, Chandawasa) including hub heights, OEM power curves, cut-in, rated, and cut-out speeds.
- **Module W2: 168-Hour Multi-Agency Hub-Height NWP Fetch & 672-Block Cubic Splines:** Ingest 168 hours of synoptic weather forecasts across 143 ensemble members from ECMWF IFS, GFS, ICON, and GEM; extract wind speed directly at turbine hub height (100m/120m/140m), surface atmospheric pressure, and 2m ambient temperature; fit continuous 168-hour natural cubic splines to reconstruct 672-block wind velocity matrix.
- **Module W3: Dynamic Atmospheric Air Density & IEC 61400-12 Density Normalization:** Evaluate hourly atmospheric air density $\rho(H)$ using ideal gas thermodynamics; interpolate air density to 672 blocks; apply IEC 61400-12 dynamic velocity scaling to normalize raw hub-height wind speeds to standard aerodynamic effective velocity.
- **Module W4: Atmospheric Boundary Layer Shear & Hellmann Diurnal Extrapolation:** Model vertical wind shear profiles using the Hellmann power law; apply diurnal shear exponent modulation between daytime convective ($\alpha = 0.12$) and nocturnal stable boundary layer regimes ($\alpha = 0.25$) where hub height differs from NWP reporting height.
- **Module W5: Jensen's-Inequality-Safe Member-by-Member Non-Linear Power Transformation:** Transform density-normalized effective wind speed for each NWP ensemble member into gross turbine active power via piecewise linear interpolation against OEM power curves BEFORE ensembling; mathematically protects against cubic non-linear expectation skew: $E[P(v)] \neq P(E[v])$.
- **Module W6: 28-Regime Diurnal Ensembling, Wind Velocity Weighting & Consensus Synthesis:** Partition the 7-day horizon into 28 operational wind regimes (4 regimes/day x 7 days); score ensemble members against medium-range loss functions; enforce mandatory agency diversity capping (maximum 2 models per NWP agency); compute Bayesian inverse-loss weights; apply cosine cross-fading; synthesize consensus gross power curve.
- **Module W7: Three-Factor Empirical Park Derating & High-Wind Cut-out Protection:** Apply master empirical park derating factor $\eta_{park} = 0.89$ (combining wake interaction 95%, electrical collector losses 97%, turbine availability 96.6%); enforce high-wind hysteresis cut-out protection if wind speed exceeds 25.0 m/s.
- **Module W8: DynamoDB 7-Day Planned Turbine Maintenance & 24-Hour Continuous Operation:** Query DynamoDB table (vedanjay-control-windows) for multi-day planned turbine maintenance across

the 7-day window; scale active turbine count and capacity; guarantee 24-hour round-the-clock wind generation with strict prohibition of night zeroing.

- **Module W9: Canonical Statutory 672-Block CSV Formatting & Amazon S3 Dispatch:** Format finalized schedule into statutory CSV files with 672 rows, continuous or resetting block index, 'From', 'To', 'SCH_MW', and 'AvC_MW'; upload to Amazon S3 data lake paths for RLDC/SLDC grid submission and operational dashboards.

3.3 Module W1: 7-Day Horizon Discretization & Wind Hardware Profile Ingestion

Module Context: Week-Ahead wind scheduling operates across a medium-range rolling 7-day planning window comprising exactly 672 continuous 15-minute dispatch intervals. The engine resolves execution timing in Indian Standard Time (IST, UTC+05:30):

$$D_{\text{today}} = \text{date}(\text{now}_{\text{IST}}), D_{\text{target}}(d) = D_{\text{today}} + d \text{ for } d \in \{1, 2, \dots, 7\}$$

Where: Dynamic 7-day calendar date array resolution in Asia/Kolkata timezone.

Wind Fleet Hardware Profile Ingestion: The engine loads site-specific hardware specifications for all registered wind plants. The following table details the engineering specifications for active wind generation fleets:

Plant Name	Location / State	Turbine OEM Model	Turbine Rating	Unit Count	Total MW	Hub Height	Rotor Diameter
JEWLI	Jewargi, Karnataka	Siemens Gamesa SG 3.6-145	3.60 MW	28 Turbines	100.8 MW	127.5 m	145.0 m
JGBPL	Madhya Pradesh	Envision EN182-5.0	5.00 MW	10 Turbines	50.0 MW	140.0 m	182.0 m
CHANDAWASA	Madhya Pradesh	Suzlon S120-2.1 / S111	2.10 MW	24 Turbines	50.4 MW	120.0 m	120.0 m

OEM Power Curve Ingestion: For each turbine model, the engine ingests certified manufacturer OEM power curve tables relating hub-height effective wind speed v to electrical active power output $P(v)$ at standard air density $\rho_0 = 1.225 \text{ kg/m}^3$. Power curves are parameterized into discrete arrays of wind speed nodes $\{v_0, v_1, \dots, v_K\}$ with 0.5 m/s resolution.

3.4 Module W2: 168-Hour Multi-Agency Hub-Height NWP Fetch & 672-Block Cubic Splines

Module Context: Wind kinetic energy follows a cubic relationship with velocity: $P = 0.5 * \rho * A * v^3$. Consequently, small errors in wind velocity are amplified cubically in power output. To achieve maximum forecast robustness over a 7-day horizon, the engine queries Open-Meteo's ensemble weather API for a 168-hour forecast window (7 days x 24 hours) across 143 ensemble members from ECMWF IFS025, ICON Seamless, GFS Seamless, and GEM.

Hub-Height Wind Speed Ingestion: The engine fetches wind speeds directly at the turbine hub height (e.g. 120m, 140m) to eliminate vertical shear extrapolation uncertainty wherever possible. In addition, hourly

surface atmospheric pressure $P_{surf}(H)$ and 2-meter ambient temperature $T_{amb}(H)$ are ingested across all 168 synoptic hours.

Continuous 168-Hour Natural Cubic Spline Interpolation: To transform 168 hourly wind speed predictions into 672 15-minute dispatch intervals without step discontinuities, the engine fits a natural cubic spline $S_m(t)$ across all 168 hourly nodes for each member m :

$$S'_m(0) = 0, S'_m(167) = 0, S_m(t) = a_i + b_i(t - t_i) + c_i(t - t_i)^2 + d_i(t - t_i)^3$$

Where: Natural cubic spline formulation fitted across 168 hourly wind speed nodes for ensemble member m .

Reconstruction of 672-Block Wind Velocity Matrix: The spline $S_m(t)$ is evaluated at the 672 block midpoints $t_{mid}(b) = (b - 0.5) * 0.25$ hours, clipped to the physical range $[0.0, 35.0]$ m/s, yielding the multi-member hub velocity matrix of shape $(M, 672)$:

$$v_{hub,m}(b) = \text{clip}(S_m(t_{mid}(b)), 0.0, 35.0) \text{ for all } m \in \{1, \dots, M\}, b \in \{1, \dots, 672\}$$

Where: Multi-member 672-block hub-height wind velocity matrix.

3.5 Module W3: Dynamic Atmospheric Air Density & IEC 61400-12 Density Normalization

Module Context: Aerodynamic wind turbine power generation depends directly on the mass flow rate of air passing through the rotor swept area. Atmospheric air density varies significantly across seasonal cycles and diurnal temperature swings, ranging from 1.12 kg/m³ in hot semi-arid summer conditions to 1.28 kg/m³ in cold winter mornings.

Thermodynamic Air Density Evaluation: Using the ideal gas equation of state, atmospheric air density $\rho(H)$ is computed at each synoptic hour H in $\{0, 1, \dots, 167\}$:

$$\rho(H) = \frac{P_{surf}(H) \times 100}{R_{spec} \times (T_{amb}(H) + 273.15)} \left[\frac{\text{kg}}{\text{m}^3} \right]$$

Where: Ideal gas thermodynamic air density calculation where P_{surf} is in hPa and T_{amb} is in Celsius.

Continuous 672-Block Density Interpolation: Hourly air density values are interpolated to all 672 15-minute dispatch blocks via linear interpolation:

$$\rho(b) = \text{interp1d}(\{\rho(H)\}_{H=0}^{167}, t_{mid}(b)) \text{ for all } b \in \{1, 2, \dots, 672\}$$

Where: Continuous 15-minute atmospheric air density vector across 672 blocks.

IEC 61400-12 Dynamic Velocity Normalization: In accordance with international wind turbine power performance standard IEC 61400-12, effective aerodynamic wind velocity $v_{eff}(b)$ is derived by normalizing the actual hub-height velocity to standard air density $\rho_0 = 1.225$ kg/m³:

$$v_{\text{eff},m}(b) = v_{\text{hub},m}(b) \times \left(\frac{\rho(b)}{\rho_0} \right)^{1/3} = v_{\text{hub},m}(b) \times \left(\frac{\rho(b)}{1.225} \right)^{1/3}$$

Where: IEC 61400-12 dynamic air density normalized effective wind velocity across 672 blocks.

3.6 Module W4: Atmospheric Boundary Layer Shear & Hellmann Diurnal Extrapolation

Module Context: When NWP reporting height z_{nwp} differs from turbine centerline hub height z_{hub} (e.g. 100m NWP feed vs. 140m hub height), the engine models vertical wind velocity profiles across the atmospheric boundary layer using the Hellmann power law:

$$v(z_{\text{hub}}, b) = v(z_{\text{nwp}}, b) \times \left(\frac{z_{\text{hub}}}{z_{\text{nwp}}} \right)^{\alpha(b)}$$

Where: Hellmann power-law vertical wind profile formulation where $\alpha(b)$ is the diurnal wind shear exponent.

Diurnal Shear Exponent Modulation: The wind shear exponent $\alpha(b)$ is not a constant; it exhibits strong diurnal variation driven by solar insolation and thermal atmospheric stability:

- **Daytime Convective Boundary Layer (09:00 to 17:00 IST):** Strong solar heating generates vigorous thermal convection and turbulent mixing, causing uniform vertical velocity profiles: α_{day} in [0.10, 0.14].
- **Nocturnal Stable Boundary Layer (19:00 to 06:00 IST):** Radiative surface cooling induces thermal temperature inversion, decoupling high-altitude gradient winds from surface friction and forming low-level nocturnal jets: α_{night} in [0.22, 0.28].

$$\alpha(b) = \alpha_{\text{day}} + \frac{\alpha_{\text{night}} - \alpha_{\text{day}}}{2} \left[1 + \cos\left(\frac{2\pi(b_{\text{day}}(b) - 12)}{96}\right) \right]$$

Where: Continuous harmonic diurnal modulation of the Hellmann wind shear exponent across 96 daily blocks.

3.7 Module W5: Jensen's-Inequality-Safe Member-by-Member Non-Linear Power Transformation

Module Context: In conventional naive forecasting architectures, ensemble wind speed predictions are averaged across members to produce a mean wind speed, which is then passed through the wind turbine power curve: $P_{\text{naive}} = P(E[v])$. This practice is mathematically flawed and introduces severe forecast underestimation error due to Jensen's Inequality.

Mathematical Proof of Jensen's Inequality Distortion: The wind turbine power curve $P(v)$ is a convex function in the partial load operating regime (between cut-in speed 3.0 m/s and rated speed 11.5 m/s), following cubic power $P(v)$ proportional to v^3 . For any strictly convex function f and non-degenerate random variable X , Jensen's inequality dictates:

$$\mathbb{E}[P(v)] > P(\mathbb{E}[v]) \text{ for } v \in [v_{\text{cut_in}}, v_{\text{rated}}]$$

Where: Jensen's inequality proof establishing systemic underestimation in naive ensemble-averaged wind speeds.

Member-by-Member Non-Linear Power Transformation: To eliminate Jensen's inequality distortion entirely, the Intellis Week-Ahead engine applies the non-linear OEM power curve $P_{\text{OEM}}(v)$ independently to each ensemble member's effective velocity vector BEFORE any ensemble averaging is performed across all 672 blocks:

$$P_{\text{gross},m}(b) = P_{\text{OEM}}(v_{\text{eff},m}(b)) \text{ for all } m \in \{1, \dots, M\}, b \in \{1, \dots, 672\}$$

Where: Member-by-member non-linear power curve transformation across all 672 continuous dispatch blocks.

Piecewise Linear OEM Power Curve Interpolation: For an effective velocity v_{eff} falling between tabulated OEM speed nodes v_k and v_{k+1} , gross electrical power is evaluated via piecewise linear interpolation:

$$P_{\text{OEM}}(v) = \begin{cases} 0.0 & \text{if } v < v_{\text{cut_in}} \text{ or } v \geq v_{\text{cut_out}} \\ P_k + \frac{P_{k+1} - P_k}{v_{k+1} - v_k} (v - v_k) & \text{if } v_k \leq v < v_{k+1} < v_{\text{rated}} \\ P_{\text{rated}} & \text{if } v_{\text{rated}} \leq v < v_{\text{cut_out}} \end{cases}$$

Where: Piecewise continuous OEM wind turbine power curve transfer function.

3.8 Module W6: 28-Regime Diurnal Ensembling, Wind Velocity Weighting & Consensus Synthesis

Module Context: Wind regimes exhibit pronounced diurnal thermodynamic periodicity. Across a 7-day medium-range horizon, the engine partitions each 24-hour day into four (4) distinct meteorological operational regimes, yielding exactly 28 operational regimes across the complete 672-block horizon:

- **Regime 1 — Nocturnal Boundary Layer Jet (R_noct):** Blocks 1 to 24 (00:00 to 06:00 IST) — Dominated by nocturnal boundary layer stability, low-level jets, and high mechanical shear.
- **Regime 2 — Morning Inversion Dissipation (R_morn):** Blocks 25 to 40 (06:00 to 10:00 IST) — Sunrise surface heating, nocturnal inversion breakup, and descending momentum transfer.
- **Regime 3 — Diurnal Convective Mixing (R_conv):** Blocks 41 to 64 (10:00 to 16:00 IST) — Vigorous convective mixing, high aerodynamic turbulence intensity, and uniform vertical velocities.
- **Regime 4 — Evening Stabilization & Jet Onset (R_stab):** Blocks 65 to 96 (16:00 to 24:00 IST) — Evening surface radiative cooling, thermal turbulence decay, and nocturnal shear re-establishment.

Global Regime Index	Horizon Day	Diurnal Phase	Local Time (IST)	Daily Block Range	Global Block Range
Regime 1	Day T+1	Nocturnal Jet	00:00 to 06:00 IST	Blocks 1 to 24	Blocks 1 to 24
Regime 2	Day T+1	Morning	06:00 to 10:00 IST	Blocks 25 to 40	Blocks 25 to 40

		Dissipation			
Regime 3	Day T+1	Diurnal Convection	10:00 to 16:00 IST	Blocks 41 to 64	Blocks 41 to 64
Regime 4	Day T+1	Evening Stabilization	16:00 to 24:00 IST	Blocks 65 to 96	Blocks 65 to 96
Regime 5 to 8	Day T+2	4 Diurnal Regimes	00:00 to 24:00 IST	Blocks 1 to 96	Blocks 97 to 192
Regime 9 to 12	Day T+3	4 Diurnal Regimes	00:00 to 24:00 IST	Blocks 1 to 96	Blocks 193 to 288
Regime 13 to 16	Day T+4	4 Diurnal Regimes	00:00 to 24:00 IST	Blocks 1 to 96	Blocks 289 to 384
Regime 17 to 20	Day T+5	4 Diurnal Regimes	00:00 to 24:00 IST	Blocks 1 to 96	Blocks 385 to 480
Regime 21 to 24	Day T+6	4 Diurnal Regimes	00:00 to 24:00 IST	Blocks 1 to 96	Blocks 481 to 576
Regime 25 to 28	Day T+7	4 Diurnal Regimes	00:00 to 24:00 IST	Blocks 1 to 96	Blocks 577 to 672

Agency Diversity Capping ($K_{\text{agency}} \leq 2$): For each of the 28 operational regimes r in $\{1, \dots, 28\}$, candidate models are selected subject to the mandatory agency diversity constraint: a maximum of two (2) models may be chosen from ECMWF, GFS, ICON, or GEM. Bayesian inverse-loss weights are calculated to synthesize regime consensus:

$$w_k(r) = \frac{\text{MAE}_k(r)^{-2}}{\sum_{j=1}^{K_{\text{slot}}} \text{MAE}_j(r)^{-2}}, P_{\text{regime}}(b) = \sum_{k=1}^{K_{\text{slot}}} w_k(r) \times P_{\text{gross},k}(b)$$

Where: Bayesian inverse-loss weighted consensus gross power within operational wind regime r .

Robust Ensemble Median Consensus Fallback: In operational deployments across medium-range horizons where historical validation data is unavailable, the engine evaluates the non-parametric median gross power curve across all M ensemble members:

$$P_{\text{gross_consensus}}(b) = \text{median}(\{P_{\text{gross},m}(b)\}_{m=1}^M) \text{ for all } b \in \{1, 2, \dots, 672\}$$

Where: Non-parametric median gross wind power consensus across M ensemble members for block b .

3.9 Module W7: Three-Factor Empirical Park Derating & High-Wind Cut-out Protection

Module Context: Turbines in utility-scale wind farms do not operate in isolation; upstream rotors extract kinetic energy, creating downstream velocity deficits and elevated turbulence (wake effects). In addition, internal electrical collection networks incur impedance losses, and mechanical turbine components experience operational unavailability and blade aerodynamic fouling.

Three-Factor Combined Park Derating Formulation: The engine applies a rigorous three-factor empirical derate multiplier $\eta_{\text{park}} = 0.890$:

$$\eta_{\text{park}} = \eta_{\text{wake}} \times \eta_{\text{elec}} \times \eta_{\text{avail}} = 0.950 \times 0.970 \times 0.966 = 0.8897 \approx 0.890$$

Where: Master three-factor empirical wind park derate formulation.

- **Wake Deficit Factor:** $\eta_{\text{wake}} = 0.950$ (5.0% wake interaction deficit based on Jensen / Park kinematic wake decay modeling across turbine spacing).
- **Electrical Collection Loss Factor:** $\eta_{\text{elec}} = 0.970$ (3.0% electrical transmission losses across 33 kV underground cables, unit transformers, and pooling substation).
- **Turbine Availability & Fouling Factor:** $\eta_{\text{avail}} = 0.966$ (3.4% turbine operational unavailability, pitch lubrication, and leading-edge blade fouling).

High-Wind Hysteresis Cut-out Protection: To safeguard turbine structural integrity against extreme aerodynamic loads, turbines trigger automatic pitch-to-feather shutdown when wind speeds exceed cut-out threshold $v_{\text{cut_out}}$ (25.0 m/s). Generation remains zeroed until wind speeds drop below re-cut-in threshold $v_{\text{re_cut_in}}$ (22.0 m/s):

$$P_{\text{net}}(b) = \begin{cases} 0.0 & \text{if } v_{\text{hub}}(b) \geq v_{\text{cut_out}} \text{ or } (P_{\text{net}}(b-1) = 0 \text{ and } v_{\text{hub}}(b) < v_{\text{re_cut_in}}) \\ \min(P_{\text{gross_consensus}}(b) \times \eta_{\text{park}}, P_{\text{CAP_WIND}}) & \text{otherwise} \end{cases}$$

Where: Net wind park generation schedule incorporating empirical park derating and high-wind hysteresis cut-out.

3.10 Module W8: DynamoDB 7-Day Planned Turbine Maintenance & 24-Hour Continuous Operation

Module Context: Wind generation fleets operate continuously twenty-four hours a day, 365 days a year. Unlike solar photovoltaic systems which produce zero power during nocturnal hours, nocturnal wind power is frequently the peak generation period due to low-level nocturnal boundary layer jets.

Strict Prohibition of Night Zeroing: Under no circumstances shall nocturnal wind generation be clamped or set to zero. Any algorithm or post-processing filter that enforces night zeroing on wind assets violates aerodynamic physics and constitutes an immediate fatal grid deviation error under IEGC 2023.

DynamoDB Multi-Day Turbine Maintenance Ingestion: The engine queries AWS DynamoDB (table vedanjay-control-windows) for scheduled turbine maintenance outages across the 7-day window $[D_{\text{today}} + 1 \dots D_{\text{today}} + 7]$. When individual turbines (e.g. Turbine WTG-07 and WTG-12 at JEWLI) are offline for gearbox inspection, active fleet capacity is scaled:

$$P_{\text{avail_wind}}(b) = P_{\text{CAP_WIND}} \times \left(\frac{N_{\text{active_turbines}}(b)}{N_{\text{total_turbines}}} \right), P_{\text{maint}}(b) = \min(P_{\text{net}}(b), P_{\text{avail_wind}}(b))$$

Where: Turbine availability scaling under planned maintenance outages for block b .

3.11 Module W9: Canonical Statutory 672-Block CSV Formatting & Amazon S3 Dispatch

Module Context: The finalized 672-block Week-Ahead wind schedule is formatted into statutory CSV structures compliant with RLDC / SLDC transmission scheduling guidelines. Each row contains block enumeration, interval timestamps, active schedule, and available capacity:

S. No	From	To	SCH_MW	AvC_MW
1	2026-10-09 00:00	2026-10-09 00:15	38.45	100.80
16	2026-10-09 03:45	2026-10-09 04:00	52.18	100.80
48	2026-10-09 11:45	2026-10-09 12:00	28.60	100.80
80	2026-10-09 19:45	2026-10-09 20:00	44.75	100.80
96	2026-10-09 23:45	2026-10-10 00:00	61.20	100.80
1 (97)	2026-10-10 00:00	2026-10-10 00:15	58.90	100.80
...	...	...	...	...
96 (672)	2026-10-15 23:45	2026-10-16 00:00	49.30	100.80

Amazon S3 Multi-Path Enterprise Dispatch: The engine serializes the 672-row schedule and executes concurrent multi-path uploads to Amazon S3:

- **Test Data Lake Path:** s3://vedanjay-schedules-test-608744602858/intellis Weekhead wind/{SITE}/{TODAY_DATE}/{PLANT}_{START_DATE}_weekahead_WA.csv
- **Production Data Lake Path:** s3://vedanjay-schedules-prod/intellis Weekhead wind/{SITE}/{TODAY_DATE}/{PLANT}_{START_DATE}_weekahead_WA.csv
- **Raw Aerodynamic Forecast Archive:** s3://vedanjay-schedules-prod/raw_archives/week_ahead_wind/{YEAR}/{MONTH}/{SITE}_{START_DATE}_WA_672_raw.json

4. Enterprise Integration, Edge Cases, Verification & Developer Implementation Guide

Section 4 codifies the enterprise integration architecture, cloud infrastructure contracts, runtime execution SLAs, comprehensive edge case failure matrices, and a step-by-step developer implementation guide. This section provides an engineering blueprint enabling a junior developer or intern to construct, test, and deploy the entire Intellis Week-Ahead scheduling codebase matching the exact production specifications.

4.1 Unified Week-Ahead Data Dictionary & Interface Contracts

The following table defines the unified data contracts, storage schemas, and interface specifications governing all Week-Ahead operational workflows across Amazon Web Services (AWS) infrastructure:

Data Entity	Storage Technology	Identifier / URI Scheme	Payload Schema & Format	Lifecycle & Access Pattern
Plant Topology	DynamoDB Table	vedanjay-plant-registry	JSON item with {plant_id, type, capacity_ac, capacity_dc, tilt, azimuth, hub_height, turbines}	Read-only at initialization; cached in Lambda runtime memory.
Planned Outages	DynamoDB Table	vedanjay-control-windows	JSON item with {site, start_time, end_time, outage_type, capacity_derate_fraction}	Queried daily for 7-day window [T+1 .. T+7]; filtered by plant ID.
NWP Weather Cache	Local Ephemeral / S3	/tmp/wa_weather_cache // raw_archives/	JSON object with 168-hour arrays for GHI, DNI, wind_speed, temp_2m, press_surf	Fetches once per run; cached for 4 hours to avoid redundant API calls.
Solar WA Schedule (Test)	Amazon S3 Object	s3://vedanjay-schedules-test-608744602858/intellis Weekhead solar/...	Statutory CSV: S. No, From, To, SCH_MW, AvC_MW (672 rows)	Written at 04:35 IST daily; retained for 90 days.
Solar WA Schedule (Prod)	Amazon S3 Object	s3://vedanjay-schedules-prod/intellis Weekhead solar/...	Statutory CSV: S. No, From, To, SCH_MW, AvC_MW (672 rows)	Written at 04:35 IST daily; permanent statutory audit trail.
Wind WA Schedule (Test)	Amazon S3 Object	s3://vedanjay-schedules-test-608744602858/intellis Weekhead wind/...	Statutory CSV: S. No, From, To, SCH_MW, AvC_MW (672 rows)	Written at 04:35 IST daily; retained for 90 days.
Wind WA Schedule (Prod)	Amazon S3 Object	s3://vedanjay-schedules-prod/intellis Weekhead wind/...	Statutory CSV: S. No, From, To, SCH_MW, AvC_MW (672 rows)	Written at 04:35 IST daily; permanent statutory audit trail.
Execution Telemetry	AWS CloudWatch Logs	/aws/lambda/intellis-weekhead-forecast-*	Structured JSON logs: {status, site, duration_ms, blocks_generated, s3_uri}	Streams in real time; retention set to 30 days.

4.2 AWS Lambda Container Execution Runtime & Performance SLA

The Week-Ahead forecasting engine is engineered for serverless execution inside AWS Lambda container images. To guarantee high availability and meet statutory dispatch deadlines, the engine enforces strict operational performance SLAs:

Performance Metric	Statutory SLA Requirement	Typical Production Benchmark	Failure Threshold / Alert Condition
Lambda Invocation Duration	< 15.0 seconds per site	4.2 to 5.8 seconds per site	Alert triggered if runtime exceeds 12.0 seconds.
Memory Utilization	< 2048 MB RAM allocation	620 to 880 MB peak memory	Alert triggered if memory exceeds 80% (1638 MB).
S3 Upload Latency	< 800 milliseconds per CSV	180 to 320 milliseconds	Exponential backoff retry triggered after 2000 ms.
Schedule Generation Deadline	Before 05:00 AM IST daily	Completed by 04:38 AM IST	Critical P1 escalation if incomplete by 05:30 AM IST.
Continuous Block Integrity	Exactly 672 blocks (zero missing)	100% block completeness	Fatal error if block count != 672 or NaN detected.
Mathematical Precision	2 decimal places (0.01 MW)	Rounded via f'{val:.2f}'	Validation check flags unrounded floating-point strings.

4.3 Ten-Phase Step-by-Step Developer & Intern Implementation Guide

This subsection details the chronological, ten-phase roadmap for building, verifying, and deploying the entire Week-Ahead scheduling codebase from scratch:

- Phase 1: Environment Initialization & Dependency Pinning:** Create dedicated Python 3.11 virtual environment. Pin core numerical and cloud libraries in requirements.txt: numpy>=1.24.0, pandas>=2.0.0, scipy>=1.10.0, pvlib>=0.10.0, boto3>=1.28.0, requests>=2.31.0. Configure AWS CLI credentials with least-privilege IAM policies for S3 PutObject and DynamoDB GetItem.
- Phase 2: Registry & Geometry Configuration Engine:** Construct plant configuration registry (plant_registry.py). Define plant metadata classes with typed fields for AC capacity, DC capacity, array tilt, azimuth, elevation, latitude, longitude, and sub-generator topologies. Implement schema validators to reject missing geocoordinates or unphysical tilt angles (>90 degrees).
- Phase 3: 672-Block Astronomical Ephemeris & Clear-Sky Module:** Implement wa_sun_trajectory_model.py. Construct DatetimeIndex of 672 15-minute intervals starting at D_target(1) 00:00 IST. Compute block midpoints (+7.5 min). Implement NREL SPA solar position functions and Ineichen clear-sky radiation models. Transpose clear-sky GHI, DNI, DHI to tilted POA irradiance using the Hay-Davies model. Implement unconditional night zeroing.
- Phase 4: 168-Hour Multi-Agency NWP Ingestion & Caching Layer:** Develop wa_nwp_fetch_ensemble.py. Implement resilient HTTP client to query Open-Meteo ensemble API for 168-hour forecast horizons across ECMWF IFS, GFS, ICON, and GEM. Implement /tmp/ disk caching with SHA-256 cache keys to prevent redundant network calls. Handle HTTP 429/500 errors with jittered exponential backoff.
- Phase 5: Multi-Day Cubic Spline Interpolation Engine:** Develop wa_spline_kt_engine.py (solar) and wa_wind_nwp_ensemble.py (wind). For solar, compute hourly clearness index Kt(H) and fit daily natural cubic splines (bc_type='natural') for each of the 7 days; evaluate at 96 daily blocks and multiply by

POA_cs. For wind, fit continuous 168-hour natural cubic splines to hub-height wind speeds and evaluate at 672 block midpoints.

- **Phase 6: Aerodynamic & Thermodynamic Power Conversion Models:** Develop `wa_power_derate_model.py` (solar) and `wa_wind_aerodynamics_engine.py` (wind). For solar, simulate cell temperatures $T_{cell}(b)$ and apply Sandia/Faiman thermal derating and inverter efficiency. For wind, compute ideal gas air density $\rho(b)$, apply IEC 61400-12 velocity normalization, and convert effective speeds to gross power via OEM curves member-by-member.
- **Phase 7: Portfolio Consensus & Agency Diversity Capping Engine:** Implement multi-member ensembling modules. Partition solar days into 21 slots and wind days into 28 regimes. Implement agency diversity capping algorithm: enforce maximum 2 models per NWP agency. Evaluate Bayesian inverse-loss weights and synthesize consensus power curves. Implement non-parametric median consensus fallback.
- **Phase 8: DynamoDB Maintenance Ingestion & Multi-Generator Allocation:** Implement DynamoDB client integration with table `vedanjay-control-windows`. Parse multi-day planned outages, mapping start/end timestamps to 672 block indices. Apply capacity scaling. For multi-generator solar sites (e.g. SHAHA, ZTRIC), allocate schedules proportionally to sub-assets and preserve dedicated columns.
- **Phase 9: Four Physical Guardrails & Safety Clamping Layer:** Implement master safety screening layer. For solar, enforce astronomical night zeroing, clear-sky envelope ceiling (105%), overcast optical floor (8%), and inverter active ramp rate filtering (10% P_{ac}). For wind, enforce high-wind hysteresis cut-out (25 m/s) and strictly prohibit night zeroing.
- **Phase 10: Canonical Statutory CSV Serialization & S3 Dispatch:** Develop `wa_formatter_s3_upload.py`. Construct 672-row statutory CSV files with required columns: S, No, From, To, SCH_MW, AvC_MW. Execute multi-path Amazon S3 uploads with MD5 checksum verification. Log execution telemetry to AWS CloudWatch.

4.4 Matrix of Ten Operational Week-Ahead Edge Cases & Automated Mitigations

Medium-range forecasting over 672 continuous blocks introduces complex operational failure modes. The following matrix specifies ten (10) critical edge cases and their automated deterministic mitigations:

Case ID	Operational Failure Scenario	Root Cause & Impact	Automated Mitigation & Fallback Strategy
WA-EC-01	NWP Weather API Total Outage	Network partition or HTTP 503 from external weather providers.	Fallback to 7-day rolling climatological K_t diurnal profile and local cached forecasts.
WA-EC-02	Extreme Ensemble Divergence	Severe synoptic disagreement (e.g. developing monsoon depression).	Activate trimmed-mean consensus (discard top/bottom 15% outliers) and clamp to historical bounds.
WA-EC-03	Extreme Wind Cut-Out Event	Hub wind speed exceeds 25.0 m/s during cyclonic storm.	Enforce automatic pitch shutdown: clamp schedule to 0.0 MW with 22.0 m/s hysteresis re-cut-in.
WA-EC-04	Severe Air Density Extremes	Intense summer heatwave ($\rho < 1.12$) or winter cold snap ($\rho > 1.28$).	Thermodynamic air density calculation updates dynamically; scales effective velocity strictly per IEC 61400-12.
WA-EC-05	DynamoDB Maintenance Shift	Planned outage dates shifted mid-week by plant operators.	Dynamic query window pulls fresh maintenance state daily at 04:30 AM IST;

			re-indexes 672 blocks.
WA-EC-06	Solar Tracking System Failure	Single-axis tracker stuck at fixed tilt during high irradiance.	Engine detects tracker alarm flag; falls back to fixed-tilt POA transposition geometry.
WA-EC-07	Timezone / Leap Year Drift	Leap day (Feb 29) or daylight saving time parsing skew.	Fixed pytz.timezone('Asia/Kolkata') enforcement; day-of-year evaluated via astronomical calendar.
WA-EC-08	S3 Upload Permission Failure	Temporary IAM credential expiry or S3 rate throttling.	3-stage exponential backoff with full jitter; local disk spooling at /tmp/pending_s3_uploads/.
WA-EC-09	Multi-Generator Missing Feeder	SCADA telemetry lost for sub-generator during allocation.	Fallback to static nameplate interconnect capacity ratio defined in plant registry.
WA-EC-10	Unscheduled Grid Back-Down	SLDC issues emergency corridor curtailment order.	Ingest emergency control window; apply uniform percentage curtailment across remaining blocks.

4.5 Architectural Certification, Regulatory Compliance & Engineering Sign-Off

This technical specification constitutes the official production engineering architecture for the Intellis Week-Ahead Renewable Scheduling System. The algorithmic formulations, physical models, and data pipeline structures codified herein strictly satisfy the statutory requirements of the Indian Electricity Grid Code (IEGC 2023), CERC Deviation Settlement Mechanism (DSM) Regulations, and State Grid Codes.

By adhering to the deterministic principles, pure atmospheric physics, and robust ensembling workflows documented in this specification, engineering teams and software developers can implement and maintain an enterprise-grade renewable energy scheduling platform operating with maximum reliability, grid compliance, and commercial value.